

COMP 5350/6350 Hands-On #5 Report_Part:1

Student Name: Eristen Forsythe

1. Executive Summary

This report covers my forensic analysis for Hands-On #5, which had two parts. The first part involved analyzing a Linux SSH authentication log file called `server_connections.log` using `grep` and regular expressions. The second part involved digging into a forensically collected Windows 10 registry image using FTK Imager and Registry Explorer. From the log file I found 520 failed login attempts across 23 unique IP addresses, with only one successful login recorded. For the registry side, I was able to identify 5 user accounts, figure out what programs run at startup, find the private IP address, and pull the run command history for the system.

Question	Answer
How many total IPv4 addresses appear in the log file?	1734
How many unique IPv4 addresses appear in the log file?	30
How many failed login attempts are recorded in the log file?	520
How many successful login attempts are recorded in the log file?	1
How many unique IPv4 addresses were associated with a successful login?	1 (119.137.62.142)
How many unique IPv4 addresses were associated with an unsuccessful login?	23
How many users and groups are associated with this system according to the Security Accounts Manager?	5 users, 21 groups
What are the names of the users associated with this registry?	Administrator, aubie, DefaultAccount, Guest, WDAGUtilityAccount
When did the user aubie last log in to the system?	2020-10-23 07:26:02 UTC
What applications are automatically started when the user logs into the system and when was the last time the autostart was run?	SecurityHealth, VMware VM3DService Process, VMware User Process, OneDrive. Last run: 2020-10-23 00:01:09
What was the private IP address associated with the system?	192.168.48.141
What are the most recently executed commands from the Windows Run command window?	(c) iexplore.exe, (b) wab.exe, (a) cmd (order: c, b, a)

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2. Investigation Methodology (General Overview)

- Evidence type (authentication log file) : Linux SSH authentication log file (syslog format)
 - File name : server_connections.log
 - File hash : MD5, SHA-1, and SHA-256 hashes verified (see 3.1)
 - Tools used for examination and why they were appropriate : grep (with -oE flag for extended regular expressions and match-only output), sort, sort -u, and wc -l. These are standard Linux command-line tools appropriate for pattern matching and counting within text-based log files.
 - Command-line analysis approach : All commands were executed in a Linux terminal directly against server_connections.log. Results were captured and recorded.
 - Regular expression methodology : The IPv4 pattern \b([0-9]{1,3}\.){3}[0-9]{1,3}\b was used. Word boundaries (\b) prevent partial matches. The grouped octet pattern ([0-9]{1,3}\.) repeats three times for the first three octets, followed by a final octet [0-9]{1,3}.
 - How total counts were obtained : grep -oE outputs one match per line; piping to wc -l counts total occurrences.
 - How unique values were extracted : Inserting sort -u before wc -l deduplicates the match list before counting.
 - Documentation approach : Screenshots showing the full command typed and the full terminal output are included in each subsection.
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3. Evidence Overview

3.1 Log File Properties

- File name: server_connections.log
- File size : 225,216 bytes (220 KB)
- Hash values (MD5 / SHA-1 / SHA-256)
MD5: 72efdaaf373b8d6c8a809cc86b2a951f
SHA-1: 0cb884679a12d100d0c527244c9838a1244c470b
SHA-256:
1e4912727fa88245113d41b16a0cd25ceadba7f931e1c406542885b91254264f

A terminal window screenshot with a dark background. The prompt is '(EristenForsythe@digitalforensics)~[~/Downloads]'. The user has entered three commands: 'md5sum server_connections.log', 'sha1sum server_connections.log', and 'sha256sum server_connections.log'. The output for each command is displayed on the next line, with the hash value highlighted in yellow. The MD5 hash is 72efdaaf373b8d6c8a809cc86b2a951f, the SHA1 hash is 0cb884679a12d100d0c527244c9838a1244c470b, and the SHA256 hash is 1e4912727fa88245113d41b16a0cd25ceadba7f931e1c406542885b91254264f.

```
(EristenForsythe@digitalforensics)~[~/Downloads]
$ md5sum server_connections.log
72efdaaf373b8d6c8a809cc86b2a951f server_connections.log
$ sha1sum server_connections.log
0cb884679a12d100d0c527244c9838a1244c470b server_connections.log
$ sha256sum server_connections.log
1e4912727fa88245113d41b16a0cd25ceadba7f931e1c406542885b91254264f server_connections.log
```

- Total number of log entries: 1,999 lines
- Log type / source: Linux SSH authentication log, syslog format, host: LabSZ

3.2 Log Content Summary

- Timestamp format observed: MMM DD HH:MM:SS (Dec 10 06:55:46)
 - Presence of IPv4 addresses: Yes, 1,734 total occurrences, 30 unique addresses
 - Presence of failed login attempts: Yes, 520 instances of 'Failed password'
 - Presence of successful login attempts: Yes, 1 instance of 'Accepted password'
 - Presence of usernames or authentication-related fields: Yes, usernames, source ports, SSH version, PAM module messages.
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4. IPv4 Address Identification

4.1 Total IPv4 Addresses

- Command used : `grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' server_connections.log | wc -l`
- Regex used : `\b([0-9]{1,3}\.){3}[0-9]{1,3}\b`
- Output : 1734
- Final count : 1,734 total IPv4 address occurrences

```
(EristenForsythe@digitalforensics)-[~/Downloads]
$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' server_connections.log | wc -l
1734
(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

4.2 Unique IPv4 Addresses

- Command used : `grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' server_connections.log | sort -u | wc -l`
- Regex used : `\b([0-9]{1,3}\.){3}[0-9]{1,3}\b`
- Output : 30
- Final unique count : 30 unique IPv4 addresses

```
(EristenForsythe@digitalforensics)-[~/Downloads]
$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' server_connections.log | sort -u | wc -l
30
(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

The command uses the same IPv4 regex pattern as Section 4.1, but inserts `sort -u` into the pipeline before `wc -l`. The `sort -u` flag sorts all extracted IP addresses alphabetically and removes duplicates, so only distinct addresses are counted. The 30 unique addresses range from Chinese and Russian networks to European and South American IP blocks.

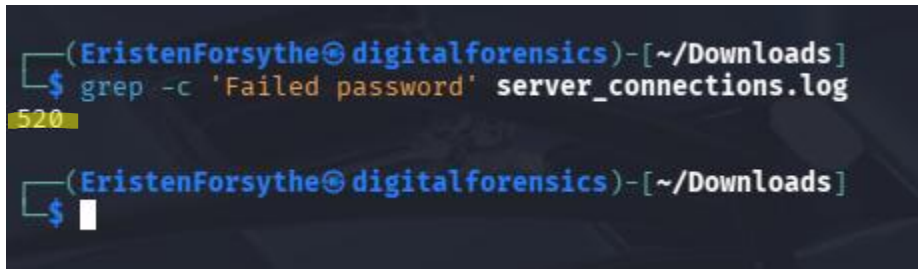
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5. Login Attempt Analysis

5.1 Failed Login Attempts

- Command used : `grep -c 'Failed password' server_connections.log`
- Output : 520
- Final count : 520 failed login attempts



```
(EristenForsythe@digitalforensics)-[~/Downloads]
$ grep -c 'Failed password' server_connections.log
520
(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

5.2 Successful Login Attempts

- Command used : `grep -c 'Accepted password\|Accepted publickey' server_connections.log`
- Output : 1
- Final count : 1 successful login attempt



```
(EristenForsythe@digitalforensics)-[~/Downloads]
$ grep -c 'Accepted password\|Accepted publickey' server_connections.log
1
(EristenForsythe@digitalforensics)-[~/Downloads]
$ grep 'Accepted password\|Accepted publickey' server_connections.log
Dec 10 09:32:20 LabSZ sshd[24680]: Accepted password for fztu from 119.137.62.142 port 49116 ssh2
```

5.3 Unique IPv4 Addresses Associated with Successful Logins

- Command used or pipeline used : `grep -E 'Accepted password|Accepted publickey' server_connections.log | grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' | sort -u | wc -l`
- Output : 1 (IP address: 119.137.62.142, associated user: fztu)
- Final unique count : 1 (IP address: 119.137.62.142, associated user: fztu)

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```
(EristenForsythe@digitalforensics) [~/Downloads]
$ grep -E 'Accepted password|Accepted publickey' server_connections.log | grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' | sort -u | wc -l
1
(EristenForsythe@digitalforensics) [~/Downloads]
$
```

```
(EristenForsythe@digitalforensics) [~/Downloads]
$ grep -E 'Accepted password|Accepted publickey' server_connections.log | grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' | sort -u
119.137.62.142
(EristenForsythe@digitalforensics) [~/Downloads]
$
```

5.3 Unique IPv4 Addresses Associated with Unsuccessful Logins

- Command used or pipeline used : `grep 'Failed password' server_connections.log | grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' | sort -u | wc -l`
- Output : 23
- Final unique count : 23 unique IP addresses associated with unsuccessful logins

```
(EristenForsythe@digitalforensics) [~/Downloads]
$ grep 'Failed password' server_connections.log | grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' | sort -u | wc -l
23
(EristenForsythe@digitalforensics) [~/Downloads]
$
```

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6. Part 1 Conclusion

- Key findings :

The log contains 1,734 IPv4 occurrences from 30 unique addresses. There were 520 failed login attempts from 23 unique IPs, and only 1 successful login (user fztu from 119.137.62.142 on Dec 10 at 09:32:20 UTC).

- Patterns observed in authentication activity :

Repeated failed attempts targeting common usernames (root, webmaster, test9, pgadmin, chen) from multiple international IPs indicate automated dictionary or brute-force attacks. Several IPs generated rapid sequential connection attempts consistent with scripted scanning tools.

- Whether the log suggests normal behavior, suspicious activity, or repeated unauthorized access attempts:

The log suggests sustained, repeated unauthorized access attempts. The volume of failed logins (520) compared to successful logins (1) is not consistent with normal behavior and is characteristic of an active brute-force campaign targeting the SSH service on host LabSZ.

- Overall forensic significance of the evidence :

This log constitutes strong evidence of a brute-force campaign against host LabSZ. The single successful login for user fztu warrants further investigation to determine whether the account credentials were compromised as a result of these attempts.

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Part 2 – Windows Registry Analysis

7. Investigation Methodology (General Overview)

- Evidence type (Windows Registry artifacts / hive files)
- File names examined : SAM, SECURITY, software, system, Users/aubie/NTUSER.DAT, Users/Default/NTUSER.DAT
- Hash values of evidence files : See Section 8.1 for MD5 hash values of each hive file
- Tools used for examination and why they were appropriate : FTK Imager was used to access and collect the protected registry hive files from the forensic archive, as it can access files that are otherwise locked by the OS. Registry Explorer was used to load and browse the offline hive files — it provides a GUI that displays keys, subkeys, values, and key last-write timestamps in a readable format. RegRipper was used to run targeted plugins against specific hives to extract key artifacts quickly using the command: `rip.pl -r <HiveFile> -p <Plugin>`.
- Manual analysis approach : Hive files were loaded into Registry Explorer one at a time. Relevant keys were navigated to directly using the known registry paths covered in class. For RunMRU entries, the RegRipper runmru plugin was also used. For logon information, the RegRipper winlogon plugin was used against NTUSER.DAT.
- Registry hives and keys examined : SAM (users and groups), SOFTWARE (autostart Run key), SYSTEM (Tcpip interface configuration), NTUSER.DAT (Winlogon key for last logon, Run key for user autostart, RunMRU for executed commands).
- How timestamps were interpreted : Registry Explorer displays the last-write timestamp of each key at the bottom of the interface. These NK cell last-write timestamps reflect the last time any value within that key was modified and were used as the primary timestamp indicator throughout this analysis.
- Documentation approach, including screenshots, annotations, and tool output : Screenshots of Registry Explorer showing the relevant key selected with its values and last-write timestamp visible are included in each section below.

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8. Evidence Overview

8.1 Registry Evidence Properties

- File name(s): SAM, SECURITY, software, system, NTUSER.DAT (from Win10Reg.7z)
- File size(s): SAM: 64 KB | SECURITY: 32 KB | software: 66 MB | system: 11 MB | NTUSER.DAT: 1.3 MB
- Hash values (MD5 / SHA-1 / SHA-256)

SAM: MD5 b91c4d579745434ab0ddc2c9a659b795

SECURITY: MD5 c1c4c56ac347f33c50a039aa5e3ef1a4

software: MD5 39053faec0fafeec8830fe05856c72a9

system: MD5 978a9bd361da672eddf7e3114888b450

NTUSER.DAT: MD5 99b7fea7c3f5124db31e853ec320c117

- Registry hives examined: SAM, SECURITY, SOFTWARE, SYSTEM, NTUSER.DAT (aubie)
-

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9. Accounts Overview: Users and Groups

9.1 Count of All Users and Groups

- Total number of users : 5 local user accounts
- Total number of groups : 21 groups (1 account group + 20 builtin aliases/security groups)
- Source Registry location(s) : SAM\Domains\Account\Users (user RID subkeys) | SAM\Domains\Account\Users\Names (account names) | SAM\Domains\Builtin\Aliases\Names (builtin groups)
- Supporting tool output : Registry Explorer, navigated to SAM hive, expanded SAM\Domains\Account\Users\Names to enumerate user accounts, and SAM\Domains\Builtin\Aliases\Names to enumerate groups. Screenshot below.

Registry Explorer v2.1.0

File Tools Options Bookmarks (2/0) View Help

Registry hives (1) Available bookmarks (2/0)

Enter text to search... Find

Key name	# values	# subkeys	Last write timestamp
C:\Users\erist\Downloads\Win1...	0	1	2020-09-08 07:51:58
ROOT	0	1	2020-09-08 07:51:58
SAM	2	3	2020-09-08 07:53:49
Domains	1	2	2020-09-08 07:51:58
Account	2	3	2020-09-08 05:58:09
Aliases	1	2	2020-09-08 07:51:58
Groups	1	2	2020-09-08 07:51:58
Users	1	6	2020-09-08 05:53:55
000001F4	3	0	2020-09-08 07:53:18
000001F5	3	0	2020-09-08 07:53:18
000001F7	4	0	2020-09-08 07:53:18
000001F8	5	0	2020-09-08 07:53:18
000003E8	4	0	2020-10-23 00:01:01
Names	1	6	2020-09-08 05:56:00
Administrator	1	0	2020-09-08 05:56:00
aubie	1	0	2020-09-08 05:53:55
DefaultAccount	1	0	2020-09-08 05:56:00
Guest	1	0	2020-09-08 05:56:00
WDAGUtilityAccount	1	0	2020-09-08 05:56:00
WDAUtilityAccount	0	0	2020-09-08 07:51:58
Builtin	2	3	2020-09-08 05:56:00
LastSkuUpgrade	1	0	2020-09-08 07:53:49
RXACT	1	0	2020-09-08 07:51:58
Associated deleted records	0	0	
Unassociated deleted values	2	0	

Values

Drag a column header here to group by that column

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Real
(default)	RegNone			☐	

Type viewer

00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26

Current offset: 0 (0x0) Bytes selected: 0 (0x0)

Key: SAM\Domains\Account\Users\Names

Selected hive: SAM Last write: 2020-09-08 05:56:00 1 of 1 values shown (100.00%) Load complete

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Registry Explorer v2.1.0

File Tools Options Bookmarks (2/0) View Help

Registry hives (1) Available bookmarks (2/0)

Enter text to search... Find

Key name	# values	# subkeys	Last write timestamp
00000227	1	0	2020-09-08 07:51:58
00000228	1	0	2020-09-08 07:51:58
0000022B	1	0	2020-09-08 07:51:58
0000022C	1	0	2020-09-08 07:51:58
0000022E	1	0	2020-09-08 07:51:58
0000022F	1	0	2020-09-08 07:51:58
00000232	1	0	2020-09-08 07:51:58
00000238	1	0	2020-09-08 07:51:58
00000239	1	0	2020-09-08 07:51:58
0000023D	1	0	2020-09-08 07:51:58
00000242	1	0	2020-09-08 07:51:58
00000243	1	0	2020-09-08 07:51:58
00000244	1	0	2020-09-08 07:51:58
00000245	1	0	2020-09-08 07:51:58
00000247	1	0	2020-09-08 07:51:58
Members	2	0	2020-09-08 07:51:58
Names	1	20	2020-09-08 05:56:00
Access Control Assistance Operators	1	0	2020-09-08 05:56:00
Administrators	1	0	2020-09-08 05:56:00
Backup Operators	1	0	2020-09-08 05:56:00
Cryptographic Operators	1	0	2020-09-08 05:56:00
Device Owners	1	0	2020-09-08 05:56:00
Distributed COM Users	1	0	2020-09-08 05:56:00
Event Log Readers	1	0	2020-09-08 05:56:00
Guests	1	0	2020-09-08 05:56:00
Hyper-V Administrators	1	0	2020-09-08 05:56:00
IIS_IUSRS	1	0	2020-09-08 05:56:00
Network Configuration Operators	1	0	2020-09-08 05:56:00
Performance Log Users	1	0	2020-09-08 05:56:00
Performance Monitor Users	1	0	2020-09-08 05:56:00
Power Users	1	0	2020-09-08 05:56:00
Remote Desktop Users	1	0	2020-09-08 05:56:00
Remote Management Users	1	0	2020-09-08 05:56:00
Replicator	1	0	2020-09-08 05:56:00
System Managed Accounts Group	1	0	2020-09-08 05:56:00
Users	1	0	2020-09-08 05:56:00
Users	0	0	2020-09-08 07:51:58
Groups	1	1	2020-09-08 07:51:58
Users	1	1	2020-09-08 07:51:58
LastSkuUpgrade	1	0	2020-09-08 07:53:49
RXACT	1	0	2020-09-08 07:51:58
Associated deleted records	0	0	
Unassociated deleted values	2	0	

Values

Drag a column header here to group by that column

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Rea
(default)	RegNone				

Type viewer

00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23

Current offset: 0 (0x0) Bytes selected: 0 (0x0)

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9.2 Names of All Users

- Names of all identified users : Administrator, aubie, DefaultAccount, Guest, WDAGUtilityAccount
- Source Registry location(s) : SAM\Domains\Account\Users\Names
- Supporting explanation and evidence : Each subkey name under the Names key corresponds directly to a local user account. The RID of each account is encoded in the key: RID 500 = Administrator, RID 501 = Guest, RID 503 = DefaultAccount, RID 504 = WDAGUtilityAccount, RID 1000 = aubie (the only non-built-in user account).

Registry Explorer v2.1.0

File Tools Options Bookmarks (2/0) View Help

Registry hives (1) Available bookmarks (2/0)

Enter text to search... Find

Key name	# values	# subkeys	Last write timestamp
C:\Users\erist\Downloads\Win1...			
ROOT	0	1	2020-09-08 07:51:58
SAM	2	3	2020-09-08 07:53:49
Domains	1	2	2020-09-08 07:51:58
Account	2	5	2020-09-08 05:56:09
Aliases	1	2	2020-09-08 07:51:58
Groups	1	2	2020-09-08 07:51:58
Users	1	6	2020-09-08 05:53:55
00000000	3	0	2020-09-08 07:53:18
00000001	3	0	2020-09-08 07:53:18
00000002	4	0	2020-09-08 07:53:18
00000003	5	0	2020-09-08 07:53:18
00000004	4	0	2020-10-23 00:01:01
Names	1	6	2020-09-08 05:56:00
Administrator	1	0	2020-09-08 05:56:00
aubie	1	0	2020-09-08 05:53:55
DefaultAccount	1	0	2020-09-08 05:56:00
Guest	1	0	2020-09-08 05:56:00
WDAGUtilityAccount	1	0	2020-09-08 05:56:00
WDAGUtilityAccount	0	0	2020-09-08 07:51:58
Buttons	2	3	2020-09-08 05:56:00
LastSkuUpgrade	1	0	2020-09-08 07:53:49
RXACT	1	0	2020-09-08 07:51:58
Associated deleted records	0	0	
Unassociated deleted values	2	0	

Values

Drag a column header here to group by that column

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated
(default)	RegName				

Type viewer

00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26

Current offset: 0 (0x0) Bytes selected: 0 (0x0) Data interpreter: ?

Key: SAM\Domains\Account\Users\Names

Selected hive: SAM Last write: 2020-09-08 05:56:00 1 of 1 values shown (100.00%) Load complete

Value: (default) Collapse all hives Hidden keys: 0

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10. Last Logon for “Aubie”

- Username: aubie
- Last logon timestamp: 2020-10-23 07:26:02
- Source Registry key(s) or artifact(s):
NTUSER.DAT\Software\Microsoft\Windows NT\CurrentVersion\Winlogon
- Supporting evidence : Per class slides (Windows Registry Forensics, Slide 35), logon information relative to when a user last logged into a system is found in NTUSER.DAT\Software\Microsoft\Windows NT\CurrentVersion\Winlogon. In Registry Explorer, this key's last-write timestamp reads 2020-10-23 07:26:02 UTC, indicating this was the last time Windows updated the Winlogon settings for user aubie, which corresponds to their last logon session.

Registry Explorer v2.1.0

File Tools Options Bookmarks (30/0) View Help

Registry hives (2) Available bookmarks (32/0)

Enter text to search... Find

Key name	# values	# subkeys	Last write timestamp
Personalization	0	1	2020-09-08 05:56:24
Phone	0	1	2020-09-08 05:56:24
Pins	0	1	2020-09-08 06:00:43
Poem	2	1	2020-10-20 05:02:36
Remote Assistance	0	0	2020-09-08 05:56:24
ScreenMagnifier	1	0	2020-09-08 05:56:24
Sensors	0	0	2020-09-08 05:56:24
SkyDrive	1	0	2020-09-08 06:01:15
Speech	0	1	2020-09-08 05:56:24
Speech Virtual	0	0	2020-09-08 05:56:24
Speech_OneCore	0	7	2020-10-21 04:12:56
SQMClient	1	0	2020-09-08 05:58:38
SystemCertificates	0	8	2020-09-08 05:57:08
TabletTP	0	1	2020-09-08 05:56:24
UEV	0	1	2020-09-08 05:56:24
Unified Store	2	1	2020-09-08 06:00:43
Unistore	1	0	2020-09-08 06:00:43
UserData	0	2	2020-10-20 05:02:33
WAB	0	1	2020-09-08 05:56:24
WinSxS	0	1	2020-09-08 05:56:24
Wscntfy	0	5	2020-09-08 05:56:24
Windows	0	7	2020-09-08 05:56:56
Windows NT	0	1	2020-09-08 05:56:24
CurrentVersion	0	14	2020-10-21 04:13:05
AppCompatFlags	0	2	2020-10-20 04:56:01
BackgroundModel	0	1	2020-09-08 05:56:24
Devices	4	0	2020-10-23 00:01:05
EFSS	0	0	2020-09-08 05:56:24
Fonts	0	0	2020-09-08 05:56:24
HostActivityManager	0	1	2020-10-23 00:01:09
ICH	0	0	2020-09-08 05:56:24
MacCorruptedFileRecovery	0	1	2020-09-08 05:56:24
PrinterPorts	4	0	2020-10-23 00:01:05
Sensor	0	1	2020-10-21 04:13:05
TileDataModel	0	3	2020-09-08 05:56:24
TokenBroker	0	1	2020-10-20 04:56:52
Windows	3	2	2020-10-20 05:05:12
Winlogon	6	0	2020-10-23 07:26:02
Windows Script Host	0	1	2020-09-08 05:56:32
Windows Search	0	1	2020-09-08 05:58:26
Windows Security Health	0	1	2020-09-08 05:58:33
Wslap	0	3	2020-09-08 05:56:24
Policies	0	2	2020-09-08 05:56:24

Values

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated
ExcludeProfileDirs	RegSz	AppData\Local\AppData\LocalLow\Recycle.B...		☐	☐
BuildNumber	RegDword	17763		☐	☐
FirstLogon	RegDword	0		☐	☐
PUUActive	RegBinary	58-98-CD-3F-01-00-01-00-06-00-0F-00-05-24...	00-64-00-65	☐	☐
DP	RegBinary	D2-00-E8-00-07-00-01-00-06-00-00-00-58-98...	7E-00-4D-00	☐	☐
ParseAutoexec	RegSz	1		☐	☐

Type viewer Binary viewer

Value name ExcludeProfileDirs

Value type RegSz

Value AppData\Local\AppData\LocalLow\Recycle.Bin\OneDrive\Work Folders

Raw value 41-00-70-00-70-00-44-00-61-00-74-00-61-00-5C-00-4C-00-6F-00-63-00-61-00-4C-00-38-00-41-00-70-00-79-00-44-00-61-00-74-00-61-00-3C-00-4C-00-6F-00-63-00-61-00-6C-00-4C-00-6F-00-77-00-38-00-24-00-52-00-65-00-63-00-79-00-63-00-8C-00-65-00-2E-00-42-00-69-00-4E-00-38-00-4F-00-4E-00-65-00-44-00-72-00-69-00-76-00-65-00-30-00-37-00-6F-00-72-00-68-00-20-00-46-00-6F-00-64-00-65-00-72-00-73-00-00-00

Selected hive: NTUSER.DAT Last write: 2020-10-23 07:26:02 6 of 6 values shown (100.00%) Load complete

Value: ExcludeProfileDirs Collapse all hives Hidden keys: 0

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Student Name: Eristen Forsythe

11. Autostart Entries and Last Run Time

11.1 Applications Started at Runtime

- List of applications started at runtime :
 - **HKLM (system-wide) —**
SOFTWARE\Microsoft\Windows\CurrentVersion\Run:
 - **SecurityHealth:** %windir%\system32\SecurityHealthSystray.exe
 - **VMware VM3DSERVICE Process:**
"C:\Windows\system32\vm3dservice.exe" -u
 - **VMware User Process:** "C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmusr
 - **HKCU (user-specific) —**
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Run:
 - **OneDrive:**
"C:\Users\aubie\AppData\Local\Microsoft\OneDrive\OneDrive.exe"
/background
- Relevant Registry key(s) : HKLM:
SOFTWARE\Microsoft\Windows\CurrentVersion\Run | HKCU:
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Run
- Supporting evidence : Registry Explorer was used to navigate to both Run keys. The HKLM key contains three entries: Windows Security Center tray icon, and two VMware Tools processes confirming the system runs inside a VMware virtual machine.

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Student Name: Eristen Forsythe

Registry Explorer v2.1.0

File Tools Options Bookmarks (32/0) View Help

Registry hives (3) Available bookmarks (4/0)

Enter text to search...

Find

Key name # values # subkeys Last write timestamp

- Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentVersion\Run
 - Run 3 0 2020-10-23 00:01:09

Values

Value Name Value Type Data Value Stack Is Deleted Data Record Reallocated

SecurityHealth	RegExpandSz	%windir%\system32\SecurityHealthSystray.exe	00-00-00-00	☐	☐
VMware VMX3DService Process	RegSz	"C:\Windows\system32\vm3dservice.exe" -u	00-00	☐	☐
VMware User Process	RegSz	"C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmusr	00-00	☐	☐

Type viewer Slack viewer Binary viewer

Value name SecurityHealth

Value type RegExpandSz

Value %windir%\system32\SecurityHealthSystray.exe

Raw value 25-00-77-00-69-00-4E-00-64-00-69-00-72-00-25-00-5C-00-73-00-79-00-73-00-74-00-65-00-6D-00-33-00-32-00-5C-00-53-00-65-00-63-00-75-00-72-00-69-00-74-00-79-00-4B-00-65-00-61-00-4C-00-74-00-68-00-53-00-79-00-73-00-74-00-72-00-61-00-79-00-2E-00-65-00-78-00-65-00-00-00

Slack 00-00-00-00

Value SecurityHealth Collapse all hives

Hidden keys: 0

Registry Explorer v2.1.0

File Tools Options Bookmarks (50/0) View Help

Registry hives (3) Available bookmarks (6/0)

Enter text to search...

Find

Key name # values # subkeys Last write timestamp

- Computer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
 - Run 1 0 2020-09-08 05:56:24

Values

Value Name Value Type Data Value Stack Is Deleted Data Record Reallocated

OneDrive	RegSz	"C:\Users\jaubie\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background	00-58-00-62-00-6F	☐	☐
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Type viewer Slack viewer Binary viewer

Value name OneDrive

Value type RegSz

Value "C:\Users\jaubie\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background

Raw value 22-00-43-00-3A-00-5C-00-55-00-73-00-65-00-72-00-73-00-6C-00-61-00-75-00-62-00-69-00-65-00-5C-00-41-00-70-00-70-00-44-00-61-00-5C-00-4C-00-4F-00-63-00-61-00-6C-00-5C-00-4D-00-69-00-63-00-72-00-4F-00-73-00-6F-00-66-00-74-00-5C-00-4F-00-6E-00-65-00-44-00-72-00-69-00-76-00-65-00-5C-00-4F-00-6E-00-65-00-44-00-72-00-69-00-76-00-65-00-2E-00-65-00-78-00-65-00-22-00-20-00-2F-00-62-00-61-00-63-00-68-00-67-00-72-00-6F-00-75-00-6E-00-64-00-00-00

Slack 00-58-00-62-00-6F

Value OneDrive Collapse all hives

Hidden keys: 0

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Student Name: Eristen Forsythe

11.2 Last Autostart Time

- Timestamp : 2020-10-23 00:01:09
- Source Registry artifact :
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run (key last-write time as shown in Registry Explorer)
- Supporting explanation : The last-write timestamp of the HKLM Run key (2020-10-23 00:01:09 UTC) represents the most recent time these autostart entries were modified, which aligns closely with the aubie last logon recorded the same day. The HKCU Run key last-write was 2020-09-08 06:01:00

Registry Explorer v2.1.0

File Tools Options Bookmarks (32/0) View Help

Registry hives (3) Available bookmarks (6/0)

Enter text to search... Find

Key name	# values	# subkeys	Last write timestamp
rempl	0	2	2020-10-20 07:03:00
RemplDemo	0	2	2020-10-20 07:03:00
Run	3	0	2020-10-23 00:01:09
RunOnce	0	0	2020-09-08 05:56:35
Search	0	0	2018-09-15 07:36:03
SecondaryAuthFactor	0	0	2018-09-15 07:34:25
SecureAssessment	3	0	2018-09-15 09:11:19
Security and Maintenance	0	1	2018-09-15 07:36:03
SettingsSync	5	7	2018-09-15 07:36:03
Setup	2	9	2020-10-20 04:54:14
SharedAccess	2	0	2018-09-15 07:36:04
SharedDLLs	23	0	2020-09-08 05:57:05
SharedPC	0	0	2018-09-15 07:36:04
Shell Extensions	0	2	2018-09-15 07:36:04
ShellCompatibility	0	5	2018-09-15 07:36:04
ShellServiceObjectDelayLoad	1	0	2018-09-15 07:36:04
SHUTDOWN	3	0	2020-10-23 00:00:40
SidelyGde	4	4	2020-10-21 04:55:02
SmartGlass	0	1	2018-09-15 07:36:04
SMDEn	0	0	2018-09-15 07:36:04
SMT	0	1	2018-09-15 07:36:04
Spectrum	6	1	2018-09-15 07:36:04
SpeechGestures	1	1	2018-09-15 07:36:04
StorageSense	1	2	2020-09-08 06:00:45
Store	0	2	2018-09-15 07:36:04
Synmgr	1	2	2018-09-15 07:36:04
SysPrepTapi	0	0	2018-09-15 07:36:04
SystemProtectedUserData	0	1	2020-09-08 05:56:23
Tablet PC	1	0	2020-09-08 05:56:25
Telephony	1	4	2018-09-15 07:36:04
ThemeManager	6	0	2020-10-23 00:00:59
Themes	5	3	2020-09-08 06:51:15
TouchKeyboard	0	1	2018-09-15 07:36:04
UFI	0	1	2020-09-08 06:00:47
Uninstall	0	17	2020-10-20 07:03:00
UpdaterHealthTools	2	0	2020-10-20 07:03:00
URL	0	2	2018-09-15 07:36:04
UserPictureChange	1	0	2018-09-15 07:34:25
UserState	0	1	2018-09-15 07:36:04
VRLProvider	1	0	2020-10-23 07:58:00
WaaSAssessment	81	1	2020-10-23 07:39:06
WebCheck	2	0	2018-09-15 07:36:04
WinBio	0	2	2018-09-15 07:36:04

Values

Drag a column header here to group by that column

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated
SecurityHealth	RegExpandSz	%windir%\system32\SecurityHealth\Systray.exe	00-00-00-00		
VMware VMXDSvcProcess	RegSz	"C:\Windows\system32\vmtoolsd.exe" -u	00-00		
VMware User Process	RegSz	"C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmtoolsd	00-00		

Type viewer Slack viewer Binary viewer

Value name: SecurityHealth

Value type: RegExpandSz

Value: %windir%\system32\SecurityHealth\Systray.exe

Raw value: 25-00-77-00-69-00-4E-00-64-00-69-00-72-00-25-00-5C-00-73-00-79-00-73-00-74-00-65-00-4D-00-33-00-32-00-5C-00-53-00-65-00-63-00-75-00-72-00-69-00-74-00-79-00-4B-00-65-00-61-00-6C-00-74-00-68-00-53-00-79-00-73-00-74-00-72-00-61-00-79-00-2E-00-65-00-78-00-65-00-00

Slack: 00-00-00-00

Key: Microsoft\Windows\CurrentVersion\Run

Selected hive: software Last write: 2020-10-23 00:01:09 3 of 3 values shown (100.00%) Load complete

Value: SecurityHealth Collapse all hives

Hidden keys: 0

Student Name: Eristen Forsythe

- Private IP address : 192.168.48.141
- Source Registry key(s) or artifact(s) :
SYSTEM\ControlSet001\Services\Tcpip\Parameters\Interfaces\{fb20c2a3-0398-4519-b978-0cfd185a45c}
- Supporting explanation : The DHCP-assigned IP address (DhcpIPAddress = 192.168.48.141), subnet mask (255.255.255.0), default gateway (192.168.48.2), and DNS server (192.168.48.2) were found in the SYSTEM hive under the Tcpip interface parameters. The 192.168.48.0/24 range is a private RFC 1918 address. The .2 gateway and DNS server address is characteristic of a VMware NAT adapter, consistent with the VMware Tools autostart entries identified in Section 11.



- Most recently executed commands :
 - [c] "C:\Program Files\internet explorer\iexplore.exe" (most recent)
 - [b] "C:\Program Files\Windows Mail\wab.exe"

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Student Name: Eristen Forsythe

- [a] cmd (least recent)
- Relevant Registry key(s) :
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU
- Order of entries if observable : MRUList value = 'cba' (read left to right = most recent to least recent)
- Supporting evidence : Analyzed using Registry Explorer navigating to the RunMRU key, and confirmed with RegRipper using the runmru plugin: rip.pl -r NTUSER.DAT -p runmru. Each entry is stored as a REG_SZ value with a single letter key. The trailing \1 character in each value is a standard Windows Explorer delimiter. The key last-write timestamp is 2020-10-21 13:38:13.

Registry Explorer v2.1.0

File Tools Options Bookmarks (30/0) View Help

Registry hives (4) Available bookmarks (08/0)

Enter text to search... Find

Key name	# values	# subkeys	Last write timestamp
AppContract	0	1	2020-10-21 13:42:46
AutoplayHandlers	1	4	2020-09-08 05:56:44
BandThrottling	0	0	2020-09-08 05:56:37
BannerSettings	1	1	2020-10-20 04:58:03
BannerStore	0	1	2020-10-23 00:01:36
BitBucket	1	1	2020-09-08 05:58:22
CabinetState	2	0	2020-10-20 04:58:03
CD Burning	2	2	2020-10-20 04:54:05
CDOpen	0	1	2020-10-21 04:08:19
CDSave	0	1	2020-10-21 13:42:49
CLSID	0	5	2020-09-08 05:56:30
ComDlg32	0	4	2020-10-21 04:08:13
Desktop	0	1	2020-09-08 06:00:46
Discardable	0	1	2020-09-08 05:56:29
DiskSpaceChecking	3	0	2020-10-22 04:24:30
ExtractionWizard	1	0	2020-10-21 06:03:01
FileCis	0	176	2020-10-21 08:44:58
HideDesktopIcons	0	1	2020-09-08 06:00:46
LogonState	2	0	2020-09-08 05:56:26
LowRegistry	0	0	2020-09-08 05:56:29
MenuOrder	0	1	2020-09-08 05:56:29
Modules	0	3	2020-10-20 04:58:05
MountPoints2	0	3	2020-10-21 13:36:29
Package Installation	1	0	2020-09-08 06:00:35
RecentDocs	18	7	2020-10-22 23:28:35
Ribbon	2	0	2020-10-22 23:30:20
RunMRU	4	0	2020-10-21 13:38:13
SearchPlatform	0	1	2020-09-08 05:56:24
Shell Folders	31	0	2020-09-08 05:56:31
Shutdown	1	0	2020-10-23 00:01:06
StartPage	2	0	2020-09-08 05:56:27
StartupApproved	0	2	2020-10-21 21:43:31
Streams	0	1	2020-09-08 05:57:44
StuckRects3	1	0	2020-10-20 08:13:00
Taskband	4	1	2020-10-22 23:30:52
TWInUI	0	1	2020-10-21 13:42:46
TypePaths	7	0	2020-10-22 03:55:19
User Shell Folders	20	0	2020-09-08 05:56:24
UserAssist	0	9	2020-09-08 05:56:28
VirtualDesktops	0	0	2020-09-08 05:56:37
VisualEffects	0	19	2020-09-08 05:56:31
Wallpapers	1	0	2020-09-08 06:07:00
WheelQuery	1	0	2020-10-23 07:27:04

Values - RunMRU

drag a column header here to group by that column

Value Name	MRU Position	Executable	Opened On
c	0	"C:\Program Files\Internet Explorer\explore.exe"	2020-10-21 13:38:13
b	1	"C:\Program Files\Windows Mail\mail.exe"	
a	2	cmd	

Total rows: 3

Type viewer Slack viewer Binary viewer

Value name: MRUList

Value type: REGSZ

Value: cba

Raw value: 63-00-62-00-61-00-00-00

Slack: A0-7B-0F-00

Key: Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU

Selected hive: NTUSER.DAT Last write: 2020-10-21 13:38:13 4 of 4 values shown (100.00%) Load complete

Value: MRUList Collapse all hives Hidden keys: 0

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Student Name: Eristen Forsythe

14. Part 2 - Conclusion

- Key findings : The registry artifacts reveal a single-user Windows 10 workstation running inside a VMware virtual machine (confirmed by VMware Tools autostart entries and 192.168.48.x NAT network configuration). The system had 5 local user accounts, with aubie (RID 1000) being the only non-built-in account.
- What user activity was identified : User aubie last logged into the system on 2020-10-23 at 07:26:02 UTC per the NTUSER.DAT Winlogon key last-write timestamp. The Run MRU entries show aubie launched Internet Explorer, Windows Address Book (wab.exe), and the command prompt via the Win+R Run dialog, last recorded on 2020-10-21 at 13:38:13.
- What system activity was identified : The HKLM Run key was last modified on 2020-10-23 at 00:01:09 UTC, indicating the system's autostart configuration was active on the day of the final logon. The system was assigned private IP 192.168.48.141 via DHCP from a VMware NAT adapter (gateway: 192.168.48.2). OneDrive was configured to launch in the background on each user logon.
- Overall forensic significance of the Registry evidence : The registry hives provide a consistent and corroborated account of system configuration and user activity. The combination of last logon time, autostart programs, network addressing, and Run MRU entries establishes a clear activity timeline for user aubie. The presence of VMware artifacts is significant for chain-of-custody documentation, confirming these hives originated from a virtualized environment.